I-PHONE CLOSET SCANNER

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REMOVING/HIDING THE EXISTING CONTENTS

- RoomPlan(Apple framework) models/classifies everything during the scan
- Puts the architecture of the closet(walls, floors, doors, windows, openings) in one category and other contents in the closet into another category
- Can see a 3D model of the room with/without the contents in the closet via a toggle button

TECHNICAL ARCHITECTURE

- RoomPlan
  - Fast, but only accurate to centimeters
- ARKit ruler
  - Accurate after filtering + calibration
  - Raw LiDAR accuracy is ± 5-10mm
  - Raycasts the LiDAR mesh from a fixed screen-center pixel
  - Applies a calibration factor from a known 12 inch reference (ruler) to cancel systematic bias

DEVELOPMENT PROCESS

- Mapped the 5 spec requirements to components before writing any code (table in README)
- Chose the architecture myself: RoomPlan for reconstruction, custom ARKit ruler for precision - RoomPlan alone can't hit 1/16"
- Used Claude Code to accelerate the Swift implementation via spec-driven prompts: the prompts specified the accuracy strategy (median-of-30 sampling, outlier rejection, calibration)
- Iterated on-device: test ⟶ find issues ⟶ refine prompt ⟶ re-test

Requirement	Where it lives	Approach
Scan the closet with cameras + sensors	**Scan** tab	Apple **RoomPlan** drives the camera + LiDAR to reconstruct room architecture in real time.
Digitally remove/hide existing contents	**Scan** tab → result	RoomPlan separates *architecture* (walls, floor, doors, windows, openings) from *objects* (clutter). The reconstruction renders both, with a **show/hide-contents toggle** to make the clutter vanish live, and an **architecture-only USDZ export** so the exported file is genuinely empty.
Calculate & display dimensions	**Scan** tab → result	Mode-adaptive: reach-in gets W × D × H (1/16" + decimal inches), area, and volume; walk-in gets a 2D floor plan, footprint area, per-wall segment lengths, ceiling min–max, and labeled doors/openings — all with the scale calibration applied.
1/16" accuracy + validation	**Ruler** + **Validation** tabs	LiDAR raycast measurement with outlier-rejected median filtering, live stability readout, and a **known-length scale calibration**; an on-device harness logs measured-vs-tape-measure error and reports % within 1/16". Full protocol in `VALIDATION.md`.
Live demo from iPhone	whole app	Native app; runs live on-device. Demo script below.

BEFORE CALIBRATION

Samples	20
Within 1/16"	95%
Mean abs error	0.020 in (0.51 mm)
RMSE	0.025 in (0.64 mm)
Max error	0.065 in (1.65 mm)
Systematic bias	-0.013 in (-0.34 mm)

timestamp	ground_truth_in	measured_in	error_in	abs_error_in	error_mm	within_1_1
2026-07-16T23:10:47Z	12	11.9697	-0.0303	0.0303	-0.771	PASS
2026-07-16T23:11:32Z	12	11.9586	-0.0414	0.0414	-1.052	PASS
2026-07-16T23:11:55Z	12	12.0239	0.0239	0.0239	0.606	PASS
2026-07-16T23:12:53Z	12	11.9934	-0.0066	0.0066	-0.168	PASS
2026-07-16T23:13:38Z	12	11.9352	-0.0648	0.0648	-1.647	FAIL
2026-07-16T23:14:13Z	12	11.9941	-0.0059	0.0059	-0.149	PASS
2026-07-16T23:14:32Z	12	12.0001	0.0001	0.0001	0.003	PASS
2026-07-16T23:14:54Z	12	12.019	0.019	0.019	0.483	PASS
2026-07-16T23:15:47Z	12	11.9723	-0.0277	0.0277	-0.704	PASS
2026-07-16T23:16:37Z	12	11.989	-0.011	0.011	-0.279	PASS
2026-07-16T23:16:59Z	12	11.9835	-0.0165	0.0165	-0.418	PASS
2026-07-16T23:18:00Z	12	11.9895	-0.0105	0.0105	-0.266	PASS
2026-07-16T23:18:55Z	12	12.0143	0.0143	0.0143	0.364	PASS
2026-07-16T23:19:25Z	12	12.0105	0.0105	0.0105	0.267	PASS
2026-07-16T23:20:44Z	12	11.9765	-0.0235	0.0235	-0.598	PASS
2026-07-16T23:21:35Z	12	11.9762	-0.0238	0.0238	-0.605	PASS
2026-07-16T23:22:04Z	12	11.9704	-0.0296	0.0296	-0.753	PASS
2026-07-16T23:22:38Z	12	12.0014	0.0014	0.0014	0.036	PASS
2026-07-16T23:23:28Z	12	11.9908	-0.0092	0.0092	-0.234	PASS
2026-07-16T23:23:52Z	12	11.9674	-0.0326	0.0326	-0.829	PASS

AFTER CALIBRATION

Samples	20
Within 1/16"	100%
Mean abs error	0.012 in (0.31 mm)
RMSE	0.014 in (0.35 mm)
Max error	0.026 in (0.66 mm)
Systematic bias	-0.002 in (-0.06 mm)

mestamp	ground_truth_in	measured_in	error_in	abs_error_in	error_mm	within_1_
026-07-16T23:53:27Z	12	11.9835	-0.0165	0.0165	-0.419	PASS
026-07-16T23:54:22Z	12	11.9741	-0.0259	0.0259	-0.659	PASS
026-07-16T23:55:00Z	12	11.9834	-0.0166	0.0166	-0.422	PASS
026-07-16T23:55:47Z	12	12.0215	0.0215	0.0215	0.547	PASS
026-07-16T23:56:18Z	12	12.0136	0.0136	0.0136	0.344	PASS
026-07-16T23:57:28Z	12	12.0079	0.0079	0.0079	0.2	PASS
026-07-16T23:58:06Z	12	12.0051	0.0051	0.0051	0.129	PASS
026-07-16T23:58:47Z	12	11.9956	-0.0044	0.0044	-0.112	PASS
026-07-16T23:59:39Z	12	11.9977	-0.0023	0.0023	-0.058	PASS
026-07-17T00:00:33Z	12	11.9828	-0.0172	0.0172	-0.436	PASS
026-07-17T00:01:00Z	12	11.9931	-0.0069	0.0069	-0.175	PASS
026-07-17T00:01:36Z	12	12.0158	0.0158	0.0158	0.402	PASS
026-07-17T00:02:04Z	12	12.0145	0.0145	0.0145	0.369	PASS
026-07-17T00:02:40Z	12	11.9932	-0.0068	0.0068	-0.172	PASS
026-07-17T00:03:27Z	12	11.9936	-0.0064	0.0064	-0.163	PASS
026-07-17T00:04:04Z	12	11.9837	-0.0163	0.0163	-0.414	PASS
026-07-17T00:04:33Z	12	11.9818	-0.0182	0.0182	-0.462	PASS
026-07-17T00:05:02Z	12	12.0021	0.0021	0.0021	0.053	PASS
026-07-17T00:05:44Z	12	12.0181	0.0181	0.0181	0.461	PASS
026-07-17T00:06:18Z	12	11.9906	-0.0094	0.0094	-0.24	PASS

ACCURACY, LIMITATION, VALIDATION

- Before Calibration: 95% of samples were within 1/16th tolerance → After Calibration: 100% of samples were within 1/16th tolerance
- Limitations: RoomPlan error (±5 cm per meter(3.28084 ft)), sample size
- Systematic Scale Error Before: $\frac{|-0.013|}{12} \times 100 = 0.108\bar{3}\%$
- Systematic Scale Error After: $\frac{|-0.002|}{12} \times 100 = 0.0167\%$
- Before calibration, the 0.1083% bias would be ~3.4 mm on a 10 ft wall
- After calibration, the 0.0167% bias would be ~0.5mm on a 10 ft wall